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(See 6.2.)

MILITARY SPECIFICATION

RADAR DATA DISTRIBUTION GROUPS AN/GPA-106( ) (RADAR SITE)

AN/GPA-107(V) (OPERATIONS SITE)

1. SCOPE

1.1 This specification covers the requirements for a Radar Data Distribution System which will be installed at Naval radar shore stations. The distribution system consists of two equipment groups, a radar site distribution equipment and an operations site equipment. The radar site distribution equipment takes video, synchro and trigger data from search radar and Identification friend or foe (IFF) sets and distributes the data to indicators and a link or cable connection for transmission to the operations site. The operations center equipment receives data from the link or cable and distributes it to the radar indicators.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-S-2335 - Synchros, 60 Cycle, 115 Volt.
MIL-E-16400 - Electronic Equipment, Naval Ship and Shore: General Specification.
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts: Preparation for Delivery of.

STANDARDS

MILITARY

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.

FSC 5840

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
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3. REQUIREMENTS

3.1 Reliability. - An extremely high degree of reliability is required for all items furnished. The use of electronic parts and electron tubes, under conservative operating conditions, such as moderate voltages and dissipations, is recommended.

3.2 General requirements. -

3.2.1 This specification covers the requirements for Radar Data Distribution Group AN/GPA-106( ) and AN/GPA-107(V) which will be installed at Naval radar shore stations. The AN/GPA-106( ) and AN/GPA-107(V) shall meet the requirements of Specification MIL-E-16400, except as otherwise specified herein.

3.2.2 Materials and parts which represent an improvement in the state of the art over material and parts covered by applicable specifications shall be used to increase reliability, reduce size and weight, and reduce the effects of heating. Approval of such materials and parts shall be in accordance with Specification MIL-E-16400. Particular attention shall be paid to the acceptable and unacceptable material requirements of Specification MIL-E-16400.

3.2.3 Enclosure. - The equipment shall be mounted in closed relay racks with a front panel having a 19 inch width. The arrangement of major assemblies in the cabinet shall permit a logical method of building 5, 10 or 15 module systems while retaining adequate ventilation throughout the cabinets. The cable termination assembly, cable equalizer and equalizer amplifiers may be mounted in a separate package.

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3.2.3.1 The package containing the cable termination assembly, equalizers and equalizer amplifiers shall be designed for bulkhead mounting. The connection with the RG-199/U link shall be located on the bottom of the package and the output terminals shall be on the top of the package.

3.2.5 Operating conditions. - The equipment shall be capable of continuous operation in accordance with class 4 requirements of Specification MIL-E-16400.

3.2.6 Power source. - The equipment shall be capable of operation from a 115 - volt, 60 cycle source and shall provide satisfactory operation with variations of plus or minus 10 percent in line voltage and plus or minus 5 percent in frequency. Momentary impairment of operation due to plus or minus 30 percent transient of 1 second duration will be permissible, but shall not cause failure of parts or prevent resumption of normal operation when the transient has ceased.

3.2.7 Size. - Each rack assembly shall not exceed 24 inches in width, 87 inches in height and 24 inches in depth.

3.2.8 Control locking devices. - The design of control locking devices shall be in accordance with Specification MIL-E-16400.

3.2.9 Connectors. - External video connections shall be made through series "BNC" connectors, except those used in the interconnecting harness which shall be of the "AN" or "BNC" series.

3.2.10 Tropicalization. - Tropicalization will not be required.

3.2.11 Blown fuse indicators. - Blown fuse indicators shall be provided.

3.2.12 Cabling. - All internal system cabling required for interconnection of assemblies shall be supplied.

3.3 Detail requirements. -

3.3.1 The radar data distribution group AN/GPA-106( ) shall be capable of interconnecting a dual channel search radar set and a dual channel IFF set to a microwave link or coaxial cables for transmission to the remote operations center. The AN/GPA-106( ) shall also provide mixing, gating, patching and

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control for one local radar indicator. Figure 1 of this specification is a functional block diagram of the AN/GPA-106( ) with the radar group connected to one channel of the associated radar and IFF sets. Two patching functions (not shown on figure 1) shall be included to provide cross patching of the AN/GPA-106( ) output and link or cable input. All video and trigger functions of each channel of the associated radar and IFF sets shall be available for local indicator patching in order to facilitate maintenance and performance monitoring. All patching shall be of the normal-through variety with provisions for patching standby modules.

3.3.2 The radar data distribution group AN/GPA-107(V) shall be capable of interconnecting a microwave link or coaxial cables with a variable quantity of radar indicator to provide display and independent control of each indicator position. The AN/GPA-107(V) shall be capable of expansion on the basis of 5-indicator modules. Each remote radar indicator will have radar video gating and video mixing available plus implementation for future selection of a maximum of four other radar systems. Each indicator will have a video gating and mixer control unit. The terminal equipment associated with each indicator shall handle the following video data from one radar system as a minimum (a) normal video, (b) Coho moving target indicator (MTI) video, (c) integrated normal video, (d) video mapping, and (e) three IFF mode videos. Each indicator will have individual decoding of IFF data. All video data to each indicator shall be supplied by one 75 ohm coaxial cable. Range marks from the AN/GPA-106( ) shall be available at the AN/GPA-107(V) video patch panel for system range calibration.

3.3.2.1 Figure 2 of this specification is a functional block diagram of the AN/GPA-107(V) with the system connected to a microwave link. Only one indicator with its associated control, range gate generator gated video amplifier-mixer complex has been shown for simplicity. The normal capacity of the video distributors shown on figure 2 would be five indicators with the associated control gating and mixing. Video patching (not shown on figure 2) shall be provided for all video and trigger distributors for inputs and outputs on an individual module basis including standby modules. All patching functions shall be the normal-through type. All standby modules in the AN/GPA-106( ) and AN/GPA-107(V) whether patched or plugged in for operation shall be in an operational status at all times with power applied.

3.3.2.2 The video switching assembly depicted on figure 2 is connected to two radar/IFF sources instead of the full capability of four sources for simplification. Video switching shall provide independent selection of the radar system inputs for each indicator position. The video switching assembly shall also, include switching of the associated radar trigger.

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3.3.3 Major assemblies. - The radar data distribution groups AN/GPA-106( ) and AN/GPA-107(V) shall consist of groups of the following major assemblies:

Major assembly	Reference paragraph
Video patch panel	3.4.1
IFF antenna coaxial switch	3.4.2
Closed 19 inch relay-rack cabinet, CY597A/G or equivalent	3.2.3
Plate voltage power supply	3.4.6
Cable termination assembly	3.4.8
Video switch	3.4.11
IFF and video select panel	3.4.4
IFF equipment base	3.4.5
Indicator light adapter panel (link only)	3.4.22
IFF antenna control unit	3.4.3
Power supply 28 V.d.c.	3.4.7
Range gate and mixer control unit	3.4.14
Trigger switching unit	3.4.12
Range gate generator	3.4.16
Gated video amplifier	3.4.16
Video distribution amplifier	3.4.18
Trigger distribution amplifier	3.4.19
Indicator switching unit (control)	3.4.13
Slaved decoder	3.4.20
Video mixer	3.4.17
Cable equalizer	3.4.9
Equalizer amplifier	3.4.10
Synchro amplifier	3.4.21
Indicator junction box	3.4.15
Test equipment	3.4.23
Cord set	3.4.24

3.3.3.1 Major assemblies for AN/GPA-106( ). - The radar distribution group AN/GPA-106( ) shall consist of the following major assemblies including standbys where indicated:

Major assembly	Quantity
Video switch	1
IFF and video select panel	1
IFF equipment base	1
Indicator light adapter panel (link only)	1

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Major assembly	Quantity
IFF antenna control unit	1
Power supply 28 V. d. c.	2-(1 standby)
Range gate and mixer control unit	1
Trigger switching unit	1
Range gate generator	1
Gated video amplifier	1
Video patch panel	As required
IFF antenna coaxial switch	3
Plate voltage power supply	As required (1 standby)
Indicator junction box	1
Cable termination assembly - Type A (cable only)	1

3. 3. 3. 2 Major assemblies for AN/GPA-107(V) five module system - link input. - The radar distribution group AN/GPA-107(V) five module system shall consist of the following major assemblies, including standbys where indicated, when assembled to provide service to five radar indicators:

Major assembly	Quantity
Video distribution amplifier	9 (2 standbys)
Trigger distribution amplifier	2 (1 standby)
Indicator switching unit (control)	1
Plate voltage power supply	As required (1 standby)
Power supply 28 V. d. c.	As required (1 standby)
Closed 19 inch relay rack cabinet	As required
Video patch panel	As required
Slaved decoder	2 (1 standby)
Range gate generator	6 (1 standby)
Gated video amplifier	6 (1 standby)
Range gate and mixer control unit	6 (1 standby)
Video mixer	6 (1 standby)
Video switch	6 (1 standby)
Test equipment	As required
Indicator junction box	5

3. 3. 3. 3 Major assemblies - AN/GPA-107(V) ten module system - link input. - The radar distribution group AN/GPA-107(V) ten module system shall

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consist of the following major assemblies, including standbys where indicated, when assembled to provide service to ten radar indicators:

Major assembly	Quantity
Video distribution amplifier	17 (3 standbys)
Trigger distribution amplifier	2 (1 standby)
Indicator switching unit (control)	2
Plate voltage power supply	As required (1 standby)
Power supply 28 V. d. c.	As required (1 standby)
Closed 19 inch relay rack cabinet	As required
Video patch panel	As required
Slaved decoder	2 (1 standby)
Range gate generator	12 (2 standbys)
Gated video amplifier	12 (2 standbys)
Range gate and mixer control unit	11 (1 standby)
Video mixer	12 (2 standbys)
Test equipment	As required
Video switch	12 (2 standbys)
Indicator junction box	10

3. 3. 3. 4 Major assemblies - AN/GPA-107(V) fifteen module system - link input. - The radar distribution group AN/GPA-107(V) fifteen module system shall consist of the following major assemblies, including standbys where indicated, when assembled to provide service to fifteen radar indicators:

Major assembly	Quantity
Video distribution amplifier	25 (4 standbys)
Trigger distribution amplifier	3 (1 standby)
Indicator switching unit (control)	3
Plate voltage power supply	As required (1 standby)
Power supply 28 V. d. c.	As required (1 standby)
Closed 19 inch relay rack cabinet	As required
Video patch panel	As required
Slaved decoder	2 (1 standby)
Range gate generator	18 (3 standbys)
Gated video amplifier	18 (3 standbys)
Range gate and mixer control unit	16 (1 standby)
Video switch	18 (3 standbys)
Indicator junction box	15
Video mixer	18 (3 standbys)
Synchro amplifier	2 (1 standby)
Test equipment	As required

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3.3.3.5 Major assemblies - AN/GPA-107(V) five module system - cable input. - The radar data distribution group AN/GPA-107(V) five module system for cable input shall consist of all the major assemblies required by 3.3.3.2 plus the following additional major assemblies:

Major assembly	Quantity
Cable equalizer	6
Equalizer amplifier	6
Synchro amplifier	2 (1 standby)
Cable termination assembly-Type B	1
Closed 19 inch relay rack	As required

3.3.3.6 Major assemblies - AN/GPA-107(V) ten module system - cable input. - The radar data distribution group AN/GPA-107(V) ten module system for cable input shall consist of all the major assemblies required by 3.3.3.3 plus the additional major assemblies required by 3.3.3.5.

3.3.3.7 Major assemblies - AN/GPA-107(V) - fifteen module system - cable input. - The radar data distribution group AN/GPA-107(V) fifteen module system for cable input shall consist of all the major units required by 3.3.3.4 plus the following additional major assemblies:

Major assembly	Quantity
Cable equalizer	6
Equalizer amplifier	6
Synchro amplifier	3 (1 standby)
Cable termination assembly-Type B	1
Closed 19 inch relay rack	As required

3.3.3.8 Major assemblies - AN/GPA-107(V) for terminating an additional radar set. - In terminating an additional radar set, additional major assemblies shall be provided in the following proportions for the size module indicated for either cable or link input.

Five module system

Major assembly	Quantity
Video distribution amplifier	9 (2 standbys)
Trigger distribution amplifier	2 (1 standby)
Indicator switching unit	1

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Five module system (Continued)

Major assembly	Quantity
Plate voltage power supply	As required (1 standby)
Power supply 28 V. d. c.	As required (1 standby)
Video patch panel	As required
Slaved decoder	2 (1 standby)
Synchro amplifier	2 (1 standby)
Cord set	1

Ten module system

Major assembly	Quantity
Video distribution amplifier	17 (3 standbys)
Trigger distribution amplifier	2 (1 standby)
Indicator switching unit (control)	2 (1 standby)
Plate voltage power supply	As required (1 standby)
Power supply 28 V. d. c.	As required (1 standby)
Video patch panel	As required
Slaved decoder	2 (1 standby)
Synchro amplifier	2 (1 standby)
Cord set	1

Fifteen module system

Major assembly	Quantity
Video distribution amplifier	25 (4 standbys)
Trigger distribution amplifier	3 (1 standby)
Indicator switching unit (control)	3
Plate voltage power supply	As required (1 standby)
Power supply 28 V. d. c.	As required (1 standby)
Video patch panel	As required
Slaved decoder	2 (1 standby)
Synchro amplifier	3 (1 standby)
Cord set	1

3.4 Performance requirements. -

3.4.1 Video patch panel. - The video patch panel assembly shall consist of the panel and patch cords as required and shall be designed to accommodate

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the basic five-indicator module system of expansion thereby limiting the basic panel assembly required to one type for all applications throughout the AN/GPA-106( ) or AN/GPA-107(V). The video patch panel shall provide normal-through patching, that is insertion of the patch cord shall interrupt the through path and result in termination in the patch cord. The jack side of the video patch panel shall be color coded in a suitable fashion to depict routing for the normal through-condition. The impedance of all patch panel circuits shall be a nominal 75 ohms. The design of the video patch panel shall be subject to the approval of the bureau or agency concerned prior to construction.

3.4.2 IFF antenna coaxial switch. - The IFF antenna coaxial switch shall be designed to provide transfer switching between the IFF antenna, a dummy load and the operate and standby IFF sets. The switch assembly shall be designed for installation in the IFF equipment base and shall be remotely controlled by the IFF and video select panel. The IFF antenna switch shall be capable of meeting the following electrical and mechanical requirements:

(a) Voltage square wave ratio (VSWR): 1.5:1 maximum at 1000 megacycles (mc.)
(b) Insertion loss: 0.5 decibel: (db) maximum at 1000 megacycles.
(c) Voltage rating: 500 volts peak.
(d) Radio frequency (rf) power: 100 watts CW at 1 kilomegacycle: (10^9) per second (kmc/s).
(e) Nominal impedance: 50 ohms
(f) Operating voltage: 105 to 120 V. a. c.
(g) Operating current: 0.4 ampere: (holding current) or less.
(h) Crosstalk reduction: Greater than 60 db.
(i) Operating life: 5000 cycles (minimum).
(j) Switching time: 30 milliseconds or less.
(k) Rf connector: Series HN.

3.4.3 IFF antenna control unit. - The IFF antenna control unit shall be designed for mounting in the IFF equipment base and shall provide the control voltage to the IFF antenna switch and properly sequenced interruption of the associated IFF set's transmitting trigger circuits to provide suitable protection during switching. The antenna control unit may be packaged with the IFF antenna switch as a single major assembly. The IFF antenna control unit shall meet the following electrical requirements:

(a) Inputs: plus 28 V. d. c. switched for channel "A" or "B" select, plus 28 V. d. c. for "B" select, and triggers to IFF equipment.
(b) Outputs: 115 V. a. c. to IFF antenna switch when "B" selected; interrupted triggers to IFF equipment.

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(c) Switching power: 115 V.a.c. at 10 amps., 28 V.d.c. at less than 0.2 amps.

(d) Timing: Interlocking of antenna switches and control circuitry interrupts IFF triggers during antenna switching operations and must not restore trigger until all antenna switches are seated.

3.4.4 IFF and video select panel. - The IFF and video select panel shall operate on either 28 V.d.c. or 115 V.a.c. channel select signals from the associated radar set. Front panel controls shall be provided to enable maintenance personnel to select either channel of radar and IFF for viewing at the local radar indicator independent of the operating video applied to the link or cable. The video select panel shall use 28 V.d.c. control signals for the major assemblies controlled.

3.4.5 IFF equipment base. - The IFF equipment base shall consist of a built-up pedestal to support two IFF equipments and to house the IFF antenna switch and the IFF antenna control unit. One dual, safety grounded convenience outlet shall be provided within. Access to internal units is to be through a hinged front panel. Cable access shall be through open bottom of base.

3.4.6 Plate voltage power supply. - Design of the AN/GPA-106( ) and AN/GPA-107(V) shall be directed toward minimizing the number of d.c. voltages required to operate the system. The plate power supply shall be designed to supply all plate and filament voltages required by a five module system with the exception of modules having internal filament supplies. Adequate regulation of the power supply outputs shall be provided to enable the system to meet performance requirements under the primary power condition of 3.2.6. The plate power supply shall be packaged as a plug-in module to facilitate rapid replacement with standby assemblies.

3.4.7 Power supply 28 V.d.c. - The 28 V.d.c. power supply shall be designed to supply all of the control voltage requirements of the AN/GPA-106( ) and AN/GPA-107(V). The power supply shall have adequate regulation to enable the system to meet performance requirements under the primary power conditions of 3.2.6. The supply shall be packaged as a plug-in module to facilitate rapid replacement with standby assemblies.

3.4.8 Cable termination assembly. - The cable termination assembly shall be designed to provide the transition from the small diameter coaxial cable used in the radar data distribution groups to type RG-199/U which will be used in connecting the radar site with the operations site for cable input installations. Two types of cable termination assemblies shall be provided.

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The type A cable termination assembly, part of the AN/GPA-106( ), shall provide for grounding of the RG-199/U sheaths. The type B cable termination assembly, part of the AN/GPA-107(V), shall be insulated to provide an ungrounded input for both sides of the RG-199/U.

3.4.9 Cable equalizer. - The cable equalizer shall be designed to compensate for high frequency attenuation of type RG-199/U coaxial cable in lengths of 1000 to 10,000 linear feet in steps of 1000 feet. Ten plug-in filter assemblies shall be supplied to provide the above steps. Interchangeability of plug-in filter units is mandatory. Terminating impedance of the cable equalizer shall be a nominal 75-ohms.

3.4.10 Equalizer amplifier. - A video amplifier shall be required to compensate for insertion loss of the cable equalizer (see 3.4.9). The amplifier frequency response shall be capable of being set to plus or minus 1 db from 100 cycles per second (c.p.s.) to 10 megacycles per second (mcps) and have a response of plus or minus 3 db from 10 mcps to 13.5 mcps. Above 13.5 mcps a smooth, gradual roll-off, with no evidence of ringing, is desired. Amplifier gain shall be essentially fixed with fixed attenuation units substituted for unused cable equalizers. An output level setting potentiometer shall be provided to adjust the output for a two-volt peak. Input and output impedance shall be a nominal 75 ohms. Amplifier noise level in the terminated output with input terminated, is to be less than 0.02 volt peak (includes hum and random noise). Amplifier pulse response shall be: 20 millimicrosecond rise-time with less than 10 percent overshoot and undershoot; drop less than 10 percent on a 1500 microsecond pulse of 70 percent duty-cycle; with an amplifier output amplitude equal (2 volts plus or minus 0.5 volt) for pulse lengths from 0.5 to 1500 microseconds. The equalizer amplifier may be packaged with the cable equalizer as a single major assembly.

3.4.11 Video switch. - The video switch shall perform the switching function between four sets of video data as determined by selection at the associated range gate and mixer control unit. The video switch shall perform the dual channel radar switching shown on figure 1 and the video switching between radar system as shown on figure 2. The video switch shall be capable of meeting the following electrical and mechanical requirements:

(a) Input. - 2-V. peak positive video inputs; 75-ohm impedance with provision for series connection of inputs.
(b) Gain. - Approximately unity.
(c) Output. - One 2-V. (plus or minus 0.5 volt) peak positive video output: rise and fall times less than 0.02 microsecond; pulse width 0.45 to 1500 microseconds and 70 percent duty cycle capability with droop and overshoot less than 10 percent of

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nominal peak amplitude; direct-coupled output; 75-ohm impedance. Maximum of 3.5 V. peak output for low duty-cycle signals.

(d) Switching. - 28-V. d.c. controls switching dual channel A and B inputs. 28-V. d.c. applied places B unit in "ON" position and A unit in "OFF" position.

(e) Bandwidth. - Plus or minus 1 db from 20 c.p.s. to 10 mcs; 3 db down at 13.5 megacycles per second (mcs).

(f) Noise. - Noise and hum output less than 0.005 volt peak.

(g) Delay. - Pulse delay from input to output less than 0.05 microseconds.

(h) Crosstalk. - More than 46 db below 2-volt peak output.

3.4.12 Trigger switching unit. - Trigger switching unit provides coordinated dual channel radar and IFF switching at the AN/GPA-106( ) by means of relays interlocked with those of the associated radar set to allow for normal alternate selection of IFF channels (sets). Control for this unit will be located at both the radar and operations site as part of the associated radar system. The trigger switching unit shall be capable of meeting the following electrical and mechanical requirements:

(a) Inputs. - Channels "A" and "B" radar trigger and "A" and "B" pre-trigger for local (radar site) indicator and IFF equipments, respectively.

(b) Controls. - None.

(c) Outputs. - Selected radar trigger and pre-triggers.

(d) Function. - To provide for remote control of trigger selection by local operator or remote control system with appropriate termination of trigger inputs.

(e) Power requirements. - Relay power obtained from 28 V. d.c. supply.

3.4.13 Indicator switching unit. - Indicator switching unit provides for remotely controlled switching of synchro and control circuits on an independent basis for each of five indicators with control slaved to video switch selection of up to four radar system inputs. The indicator switching unit shall be designed for interconnection with additional switching units for service to 10 or 15 radar indicators and shall be capable of meeting the following electrical and mechanical requirements:

(a) Inputs. - Antenna azimuth 1X and 36X speed. System selected control voltages.

(b) Outputs. - Selected antenna azimuth, antenna azimuth to radar mapping equipment.

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(c) Power. - 115 volts, 60 c.p.s.
(d) Indicators. - Five 28 V. incandescent indicators to indicate system selected.
(e) Fuses. - 115 V. primary power (neon indicating hold), each synchro signal lead (115 V. neon indicating holder), and each system select 5 relay power (28 V. incandescent indicator).

3.4.14 Range gate and mixer control unit. - The range gate and mixer control unit shall be designed as a separate package suitable for mounting on the side of a radar indicator cabinet. All controls shall be located on an edge lighted front panel. The control unit shall provide the operator with control of the range gate generator, gated video amplifier and the mixer. The following controls shall be provided:

(a) Three video gain controls for the input of the gated video amplifier for normal video, MTI video and integrated video.
(b) Three video gain controls for the input of the mixer for gated radar video, map video and normal background video.
(c) One gate intensifier switch, momentary action, to allow the operator to determine the relative range settings of the gate controls.
(d) One range gate delay control to position the start of the second gated video (such as, the length of the first gate).
(e) One range gate width control to govern the width of the second gate.
(f) One system select switch to feed any one of four appropriately terminated radar inputs (control of video switch).

3.4.15 Indicator junction box. - The indicator junction box shall be designed as a separate package suitable for deck or bulkhead mounting. The junction box shall be designed to provide quick disconnect of flexible cables to the associated radar indicator and range gate mixer control unit as shown on figure 2. The junction box shall also supply the 6.3 V. a.c. for illumination of the range gate and mixer control unit.

3.4.16 Range gate generator and gated video amplifier. - The range gate generator shall provide for adjustable time sequenced video gates, within one radar trigger interval, for three synchronized video inputs with individual remote control of gate length and video levels. The range gate generator and gated video amplifier shall be remotely controlled by the range gate and mixer control unit and shall be capable of meeting the following electrical requirements:

(a) Inputs. - Three individual synchronized radar video inputs, (2 V. peak positive), gate delay, gate width, and three individual video gain controls remotely located.

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(b) Outputs. - One composite video (2-V. peak positive plus or minus 0.5 V.) with time-sequential gating determined by gate delay and width controls.

(c) Gating. - Provides (a) video from approximately 5 to 100 miles (b) video from approximately 5 to 150 miles; (c) video from approximately 5 to 228 miles at radar pulse repetition frequency (prf) of 360 pulse per second (p/s). Normal operation of these controls will be; (a) video from 5 to 50 miles, (b) video from 50 to 120 miles, and (c) video from 120 miles to 228 miles. Residual output gate less than 10 percent of peak video output.

(d) Bandwidth. - Plus or minus 3 db from 20 c.p.s. to 5 mc.

(e) Noise. - Less than 0.015 V. peak to peak output.

(f) Crosstalk. - Greater than 40 db below 2 V. peak output.

3.4.17 Video mixer. - The video mixer shall provide for three synchronized video data inputs with individual remotely-controlled video levels. The video mixer shall also provide for background display of normal radar video. Control for the video mixer shall be provided at the range gate and mixer control unit. The gated video amplifier, range gate generator and video mixer may be packaged as a single major assembly or two major assemblies in lieu of three. The video mixer shall be capable of meeting the following electrical requirements:

(a) Inputs. - Three synchronized video signals each of which has the following characteristics; 2 V. peak positive pulses; 75-ohm impedance; 0.45 to 1500 microseconds pulse width and 70 percent duty cycle capability with less than 10 percent overshoot or droop. Input circuit permits local or external termination for string connection.

(b) Controls. - Remote controls provide smooth and continuously variable mixing of each of the three inputs.

(c) Bandwidth. - Plus or minus 1-db from 20 c.p.s. to 10 mcs., 3-db down at 13.5 mcs.

(d) Noise. - Noise and hum output less than 0.005 V. peak.

(e) Crosstalk. - Greater than 46 db isolation from unused channels.

(f) Attenuation. - Maximum signal attenuation 46 db (each channel).

(g) Delay. - Pulse delay from input to output less than 0.05 microsecond.

(h) Output. - 2 V. positive peak pulse, limited to 3.5 V. positive peak pulse for additive inputs; 75-ohm impedance; rise and fall

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times less than 0.02 microsecond; pulse width 0.45 to 1500 microseconds and 70 percent duty cycle capability with droop and overshoot less than 10 percent of nominal peak amplitude.

3.4.18 Video distribution amplifier. - The video distribution amplifier shall provide five independent video outputs with the provision for series connection of amplifier inputs. The video distribution amplifier shall be capable of meeting the following electrical and mechanical requirements:

(a) Input. - Single 2 V. peak positive video 75-ohm impedance.
(b) Gain. - Approximately unity.
(c) Output. - Five 2 V. plus or minus 0.5 peak positive video outputs; rise and fall times of less than 0.02 microsecond; pulse width 0.45 to 1500 microseconds and 70 percent duty cycle capability with droop and overshoot less than 10 percent of nominal peak amplitude; direct-coupled outputs individually adjustable from 0 to 2.5 V.; 75-ohm impedance.
(d) Bandwidth. - Plus or minus 1-db from 20 c.p.s. to 10 mcs.; not more than 3 db down at 13.5 mc.
(e) Noise. - Noise and hum output less than 0.005 V. peak.
(f) Adjustments. - Each of five video outputs individually adjustable by slotted, locking-shaft, screwdriver adjustments.
(g) Delay. - Pulse delay from input to output less than 0.05 microsecond.

3.4.19 Trigger distribution amplifier. - The trigger distribution amplifier shall provide twelve individual trigger outputs from a single trigger input. Delay circuits shall be provided to permit adjustment of system zero range. The trigger distribution amplifier shall be capable of meeting the following electrical requirements:

(a) Input. - 20 to 50 V. peak positive pulse; pulse width 0.5 microsecond minimum 75-ohm impedance; prf - 100 to 2000 c.p.s.
(b) Gain. - Approximately unity.
(c) Output. - Twelve fixed positive outputs of approximately 30 V. peak; pulse width 3 to 5 microseconds; 75-ohm impedance; each trigger output can be shorted without adverse effect on the remaining triggers.
(d) Noise. - Less than 0.05 V. noise or hum output.
(e) Delay. - Inherent 0.4 microsecond; additional delay of 2.5 microseconds in steps of 0.05 microsecond by delay line, plus a coarse adjustment of 2.5 to 45 microseconds.
(f) Jitter. - Within the nominal range of inputs, jitter shall be less than 0.005 microsecond.

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3.4.20 Slaved decoder. - The slaved decoder shall be capable of accepting unseparated, time-shared, 3 mode video and providing the 3 separate mode video outputs synchronized with the transmitted mode. The decoder shall include front-panel controls for individual mode enabling. Video bandwidth shall have an upper frequency of 13.5 mc at the 3 db point and flat within plus or minus 1 db up to 10 mc. The low frequency response shall be sufficient to pass the highest duty-cycle selective identification feature (S.I.F.) pulse train with less than 10 percent droop of pulse train peak amplitudes. Residual gate level less than 10 percent of nominal 2-V. peak video pulse output. Cross-talk between mode video channels, measured across 75-ohm termination with unused channels terminated in 75-ohms, is to be less than 30 db below nominal 2 V. peak video output. Inputs are nominally 0.2 to 0.5 V. peak positive video and radar pre-trigger nominally 5 to 50 V. positive peak. Indicator lights shall be located on the front panel to provide repeat-back information at the AN/GPA-107(V) to indicate that the mode selected is being fed to the transmitter. All input and output video connections are to be through "BNC" coaxial connectors.

3.4.21 Synchro amplifier. - The synchro amplifier shall be capable of driving 15 type 1CT synchro control transformers. The synchro amplifier shall provide rotor power of 150 watts, 115 V., rms, 60 cycles. The input and stator output characteristics shall be as follows:

(a) Input. - One-speed or 36-speed, 90-V., 60-cycle, 3-wire ungrounded synchro stator voltages; maximum loading of signal supply equivalent to one 1CT; rotation rates up to 60 revolutions per minute (rpm) shall be accommodated.

(b) Output. - Equal to input (unity gain plus or minus 1 percent); source impedance is less than 2-ohms; power handling capacity 10 watts (will drive 15 type 1CT control transformers power-factor corrected conforming to Specification MIL-S-2335); may be used with fewer 1CT's with appropriate loading (indicator switching unit shall supply either a dummy load or a radar indicator thus providing appropriate load at all times).

(c) Construction. - Each amplifier shall use plug-in construction to permit rapid replacement of units in case of failure.

(d) Amplifier. - Open loop gain of about 1000; net gain is unity.

(e) Noise and hum. - At least 40 db below rated 60 cycle output voltage of 90 V. rms. with amplifier input terminals grounded.

3.4.22 Indicator light adaptor panel. - The indicator light adaptor panel performs the conversion of 115 V., 60 cycle indicator light repeat-back to dry-contact signals required for systems utilizing microwave input. The adaptor panel also provided for over-all system test of the indicator light

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repeat back data. The indicator light adaptor panel shall be capable of meeting the following electrical and mechanical characteristics:

(a) Inputs. - Twenty 115 V., 60 cycle on-off potentials to operate relays (115 V. a.c. or 28 V. d.c.) plus one dry-contact input to check indicator lights throughout link system.
(b) Outputs. - Twenty dry-contact signals to operate link control circuits.
(c) Power. - 28 V. d.c. for system check.
(d) Connections. - All inputs and outputs through MS-type connectors.
(e) Relays. - 28 V. d.c./115 V., 60 c.p.s. plug-in type relays.

3.4.23 Test equipment. - A test equipment system shall be provided with the radar data distribution groups with provisions for installation in the equipment cabinet. The AN/GPA-106( ) shall include one rack mounted sweep generator equivalent to the Kay-Markasweep modified for combined sweep plus markers at the sweep coaxial output. The AN/GPA-107(V) shall include one rack mounted oscilloscope equivalent to Tektronix Model RM-45A with one high gain, one wide band and one dual trace pre-amplifiers. Two probes and storage space for pre-amplifiers shall be included. Approval by the bureau or agency concerned shall be required for use of commercial test equipment other than those specified above.

3.4.24 Cord sets. - All cord sets for sets for interconnection of relay rack cabinets shall be supplied to facilitate ease of installation and standardization of equipment groups. This shall include cord sets required for the ten and fifteen module groups of the AN/GPA-107(V). Design and layout of interconnections between racks of the five module groups shall be predicated on ease of installation with particular emphasis on future system expansion.

3.5 Special tools. - The need for special tools for adjustment and servicing of the equipment shall be minimized. If used, the special tools shall be located within the cabinet in a conveniently accessible location. Special tools are defined as those tools not listed in the Federal Supply Catalog. (Copies of this catalog may be consulted in the office of the Government inspector.)

3.6 Workmanship. - Workmanship shall be in accordance with Specification MIL-E-16400.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance

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of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction inspection.- Preproduction inspection shall be conducted on prototype major assemblies and shall consist of the following tests conducted in accordance with Specification MIL-E-16400 except for the accelerated life test cycle which shall be repeated without interruption for a period of 48 hours (2 complete days), the electrical tests of 4.5 herein and any other test deemed necessary to determine conformance with the requirements of this specification:

Inspection
(a) Surface examination
(b) Operating test
(c) Weight and dimensions
(d) Supply line voltage and frequency
(e) Heat test
(f) Temperature
(g) Humidity
(h) Power
(i) Radio interference and undesired radiation
(j) Controls and control circuits
(k) Accelerated life test

4.3 Production inspection. - Each major assembly shall be subjected to the production inspection of Specification MIL-E-16400, the electrical test of 4.5, herein and any other test deemed necessary to determine conformance with the requirements of this specification.

4.4 Production control inspection. - The first equipment and one out of each seven thereafter, and the last equipment shall be subjected to the production control inspection of Specification MIL-E-16400.

4.5 Electrical tests. - The line voltage shall be varied from 20 percent below to 20 percent above nominal for a total period of 5 minutes without damage to the equipment.

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5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of shipboard repair parts.

5.1.1 Basic equipment or item.-

5.1.1.1 Preservation and packaging.- Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation and may conform to the suppliers commercial practice when such meets these requirements.

5.1.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers commercial practice when such meets these requirements.

5.1.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.2 Domestic shipment and storage or overseas shipment.- The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made (see 6.1).

5.2.1.1 Preservation, packaging, packing and marking.- The equipment, accessories, technical publications and repair parts shall be preserved and packaged by level A or C as specified (see 6.1); packed by level A or B as specified (see 6.1) and marked, in accordance with Specification MIL-E-17555.)

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6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Preservation, packaging, packing or marking requirements other than those required by 5.1 (see 5.2).

6.2 Superseding data.- This specification covers the requirements formerly covered by Specification MIL-R-22982(SHIPS), 16 August 1961.

Preparing activity:
Navy - SH
(Project 5840-N486Sh)

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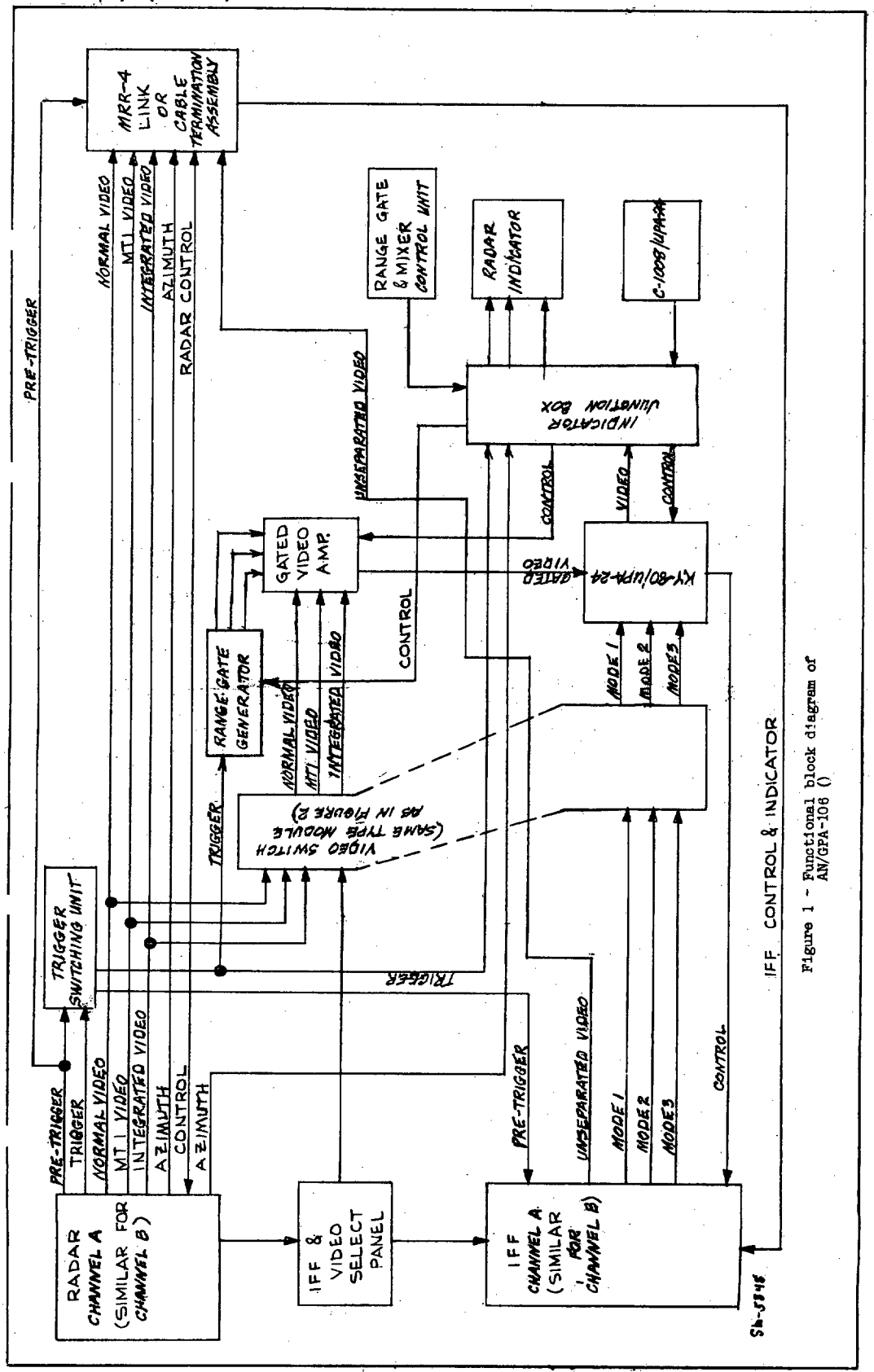


Figure 1 - Functional block diagram of AN/GPA-106 ()

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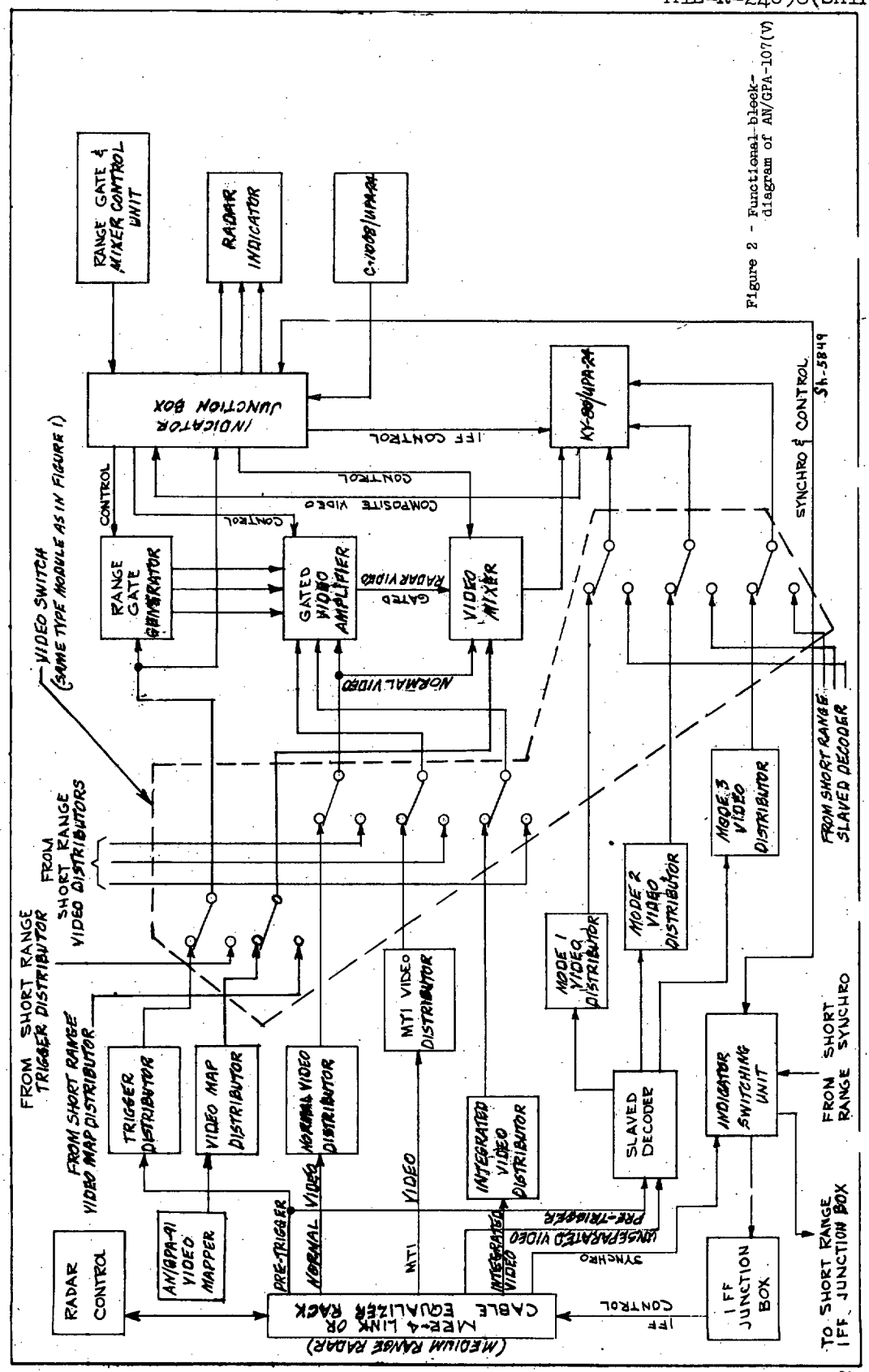


Figure 2 - Functional block-diagram of AN/GPA-107(V)

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4862, WHICH IS OBSOLETE

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